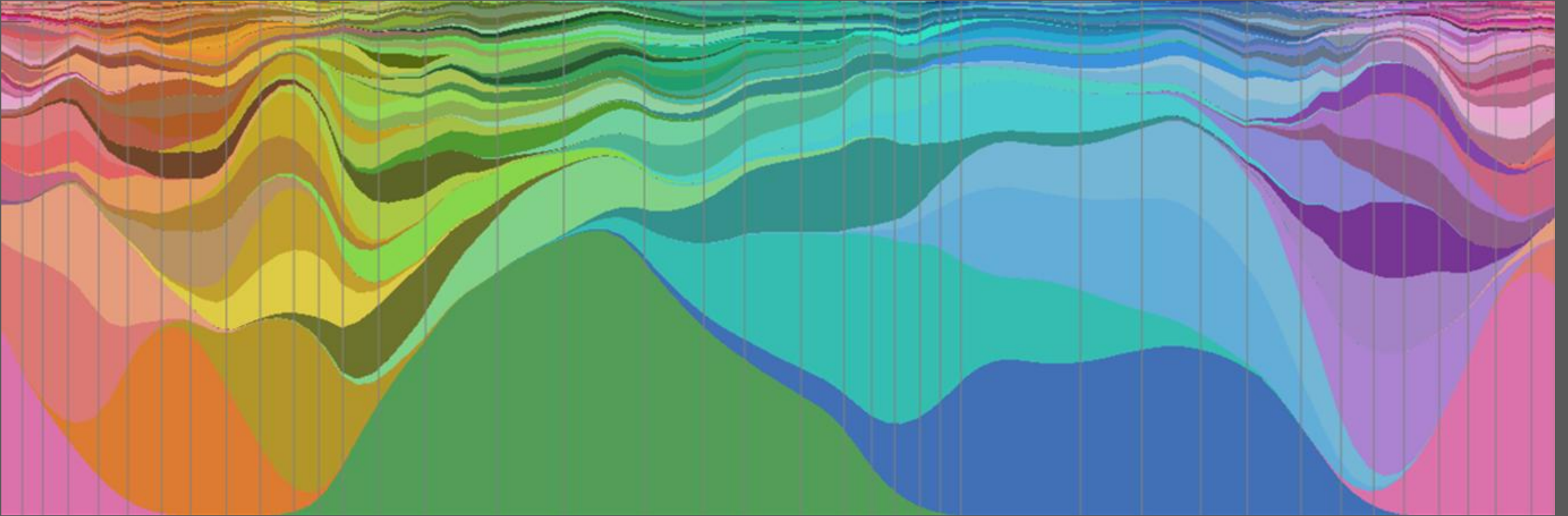


LDSCI5209 - Information Presentation and Visualisation

Week 4: Colour

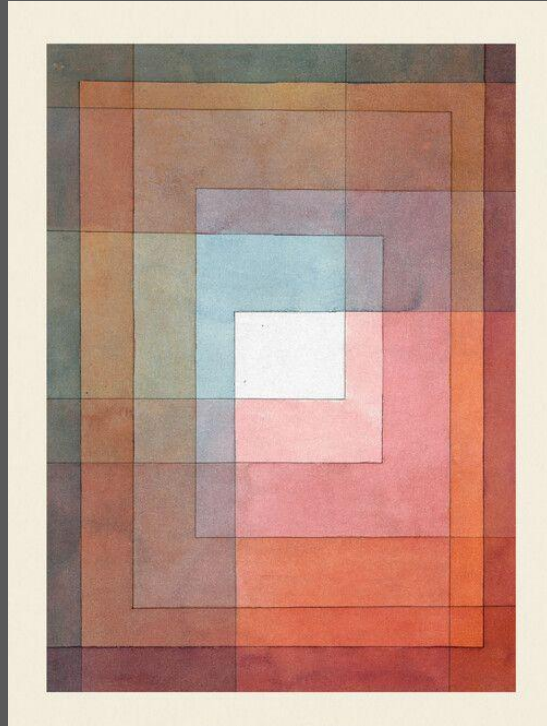


Dimitris Mylonas

Northeastern University London

Recap: The road to abstraction

- **Data & Task Abstraction:** Translate real-world data into a structured visualisation vocabulary.
- **Nested Model:** Domain, Abstraction, Visual Encoding, Algorithm.
- **Dataset Types:** Tables, networks, fields, geometry.
- **Key Visualisation Techniques:** Bar charts, scatter plots, heatmaps, line plots, parallel coordinates, flow and polar charts.



“Colour is the place where our brain and
the universe meet”

Paul Klee (1879 - 1940)

Plan for today

- Perceptual and cognitive aspects of colour
- Colour encodings
- Dimensions of colour and colour systems (interactive)

Name the danger?



Name the danger?



<https://wildlifesos.org/our-work/tigers/>

Name the fruits?



Stockman & Brainard (2009)

Name the fruits?



Stockman & Brainard (2009)

Name the fruits?



Stockman & Brainard (2009)

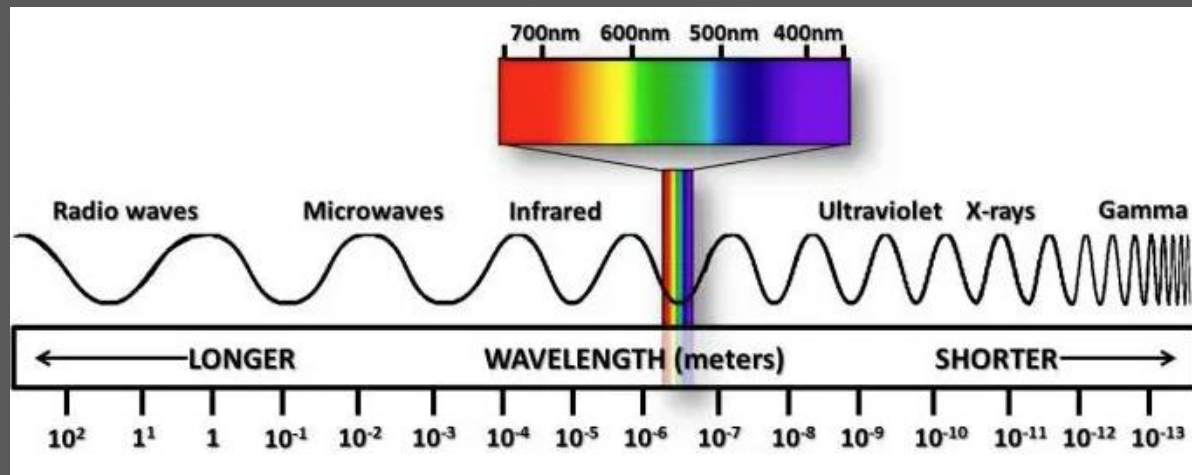
Why colour in data visualisation?

- To separate different data
- To show patterns easily
- To add meaning and emotion
- To make data clearer
- To capture attention, evoke desire and strengthen storytelling

Colour perception to colour cognition

Light, human visual system, appearance, naming

Light

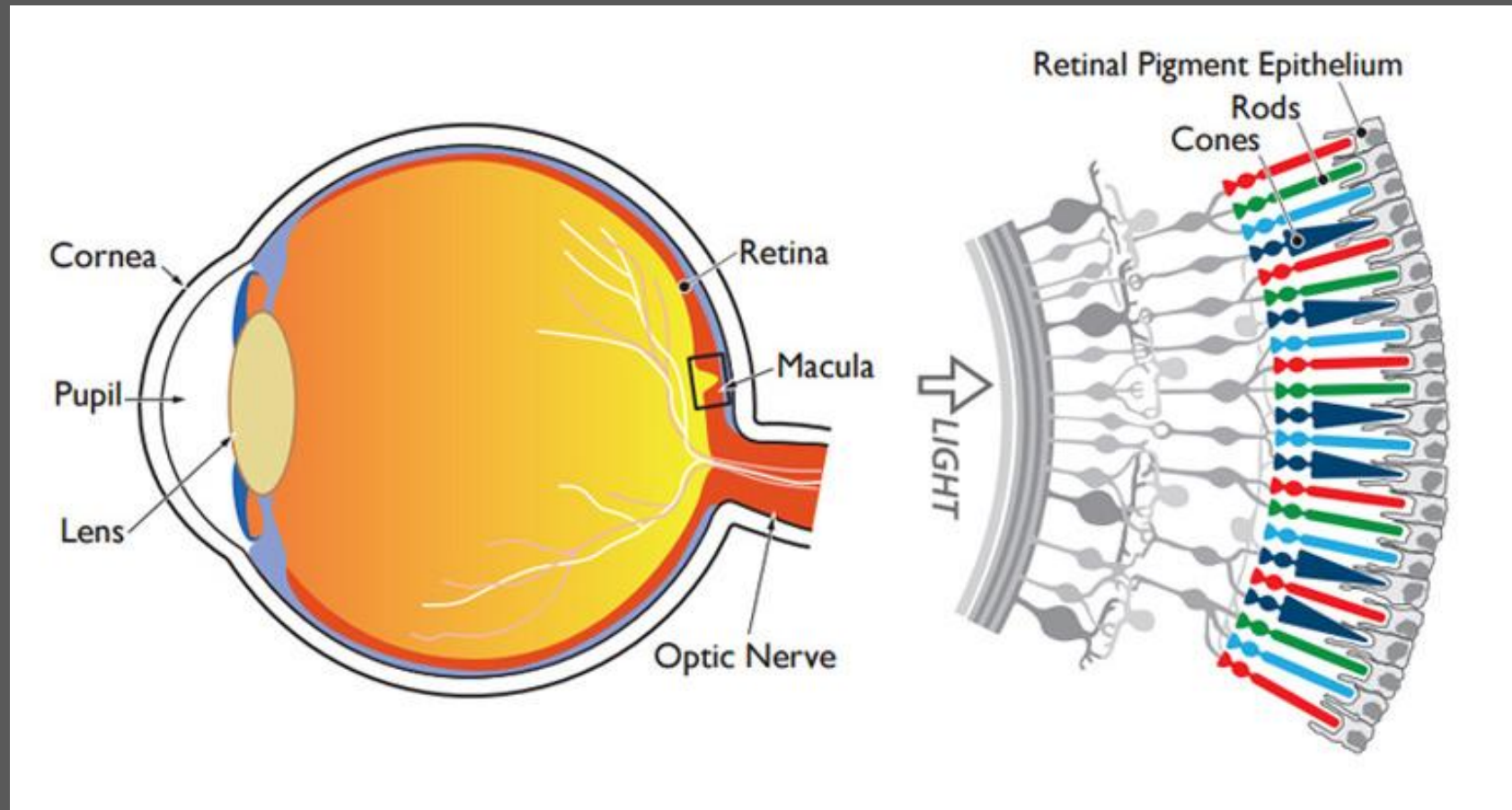


Electromagnetic spectrum



Isaac Newton (1665)

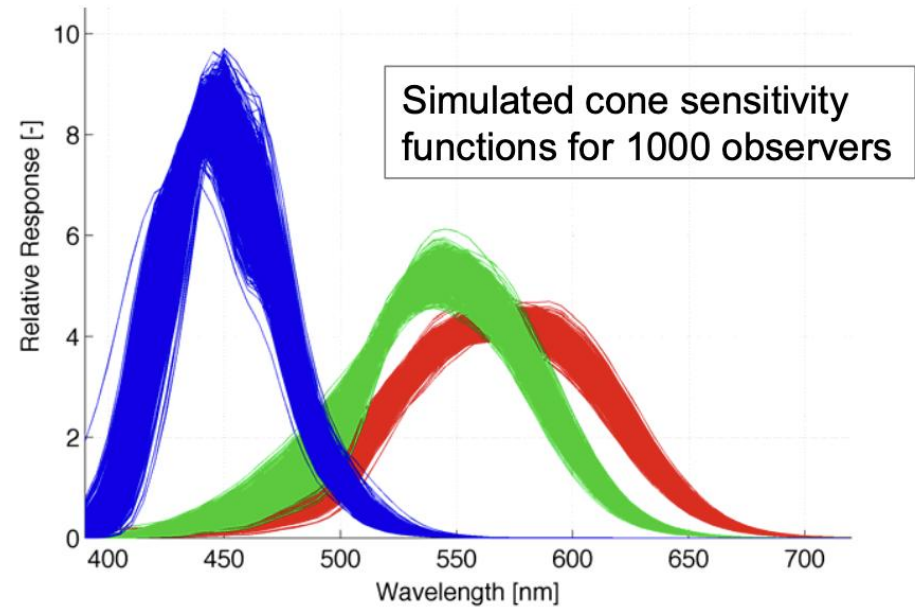
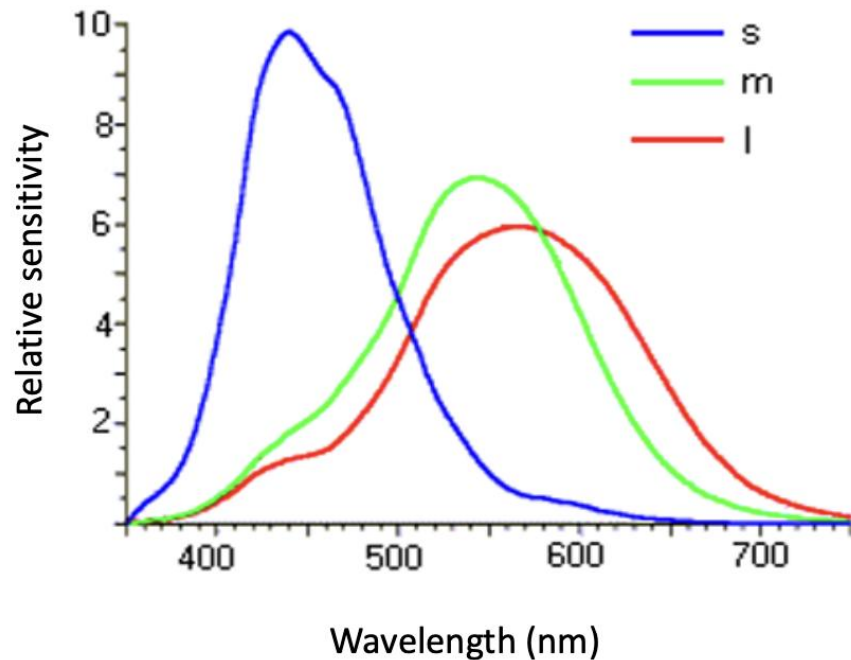
Human eye and the retina



Rods: ~125 million
Cones: ~6-7 million

Cone photoreceptors

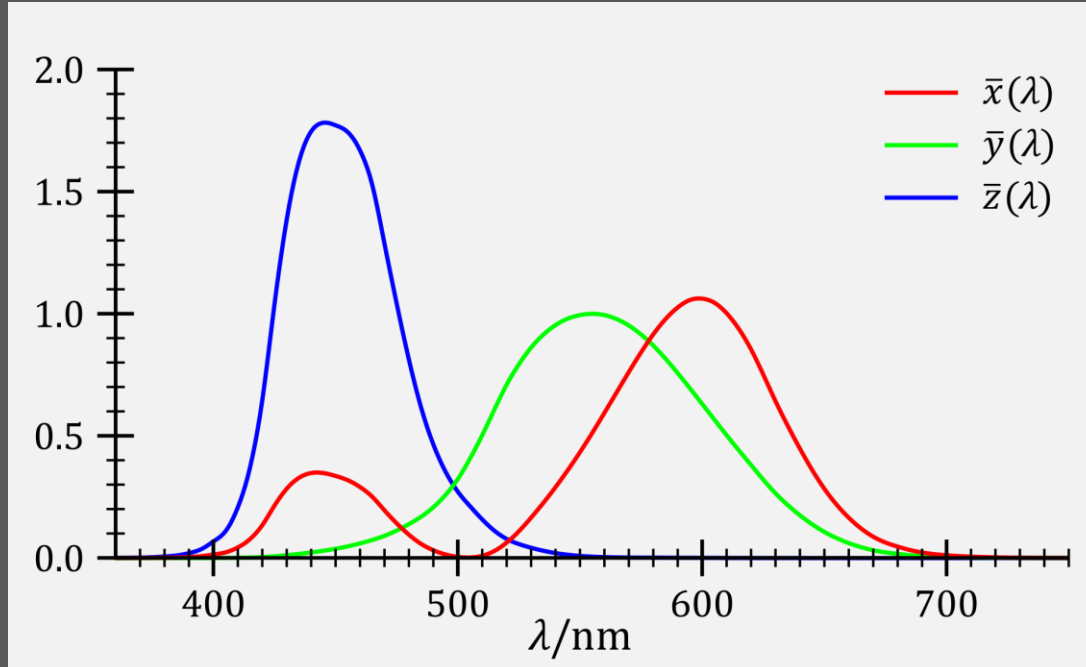
trichromacy = three independent channels for processing colour information



www.rit.edu/cos/colorscience/re_AsanoObserverFunctions.php

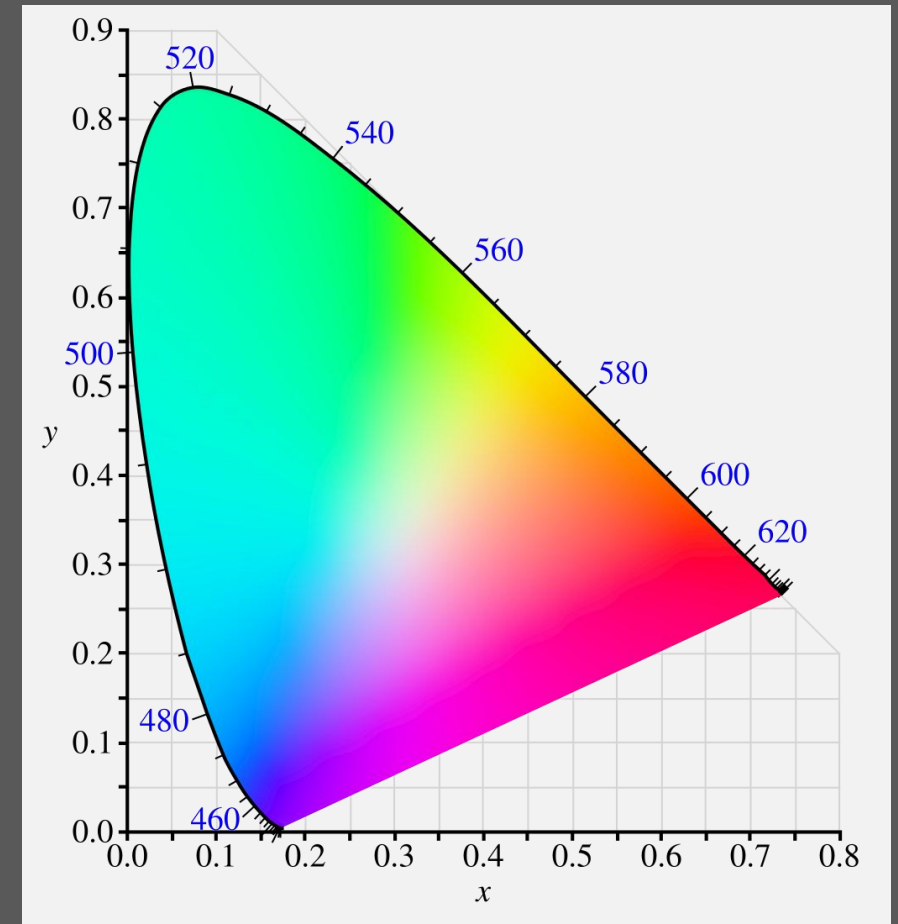
Stockman & Sharpe (2000)

CIE XYZ colour spaces

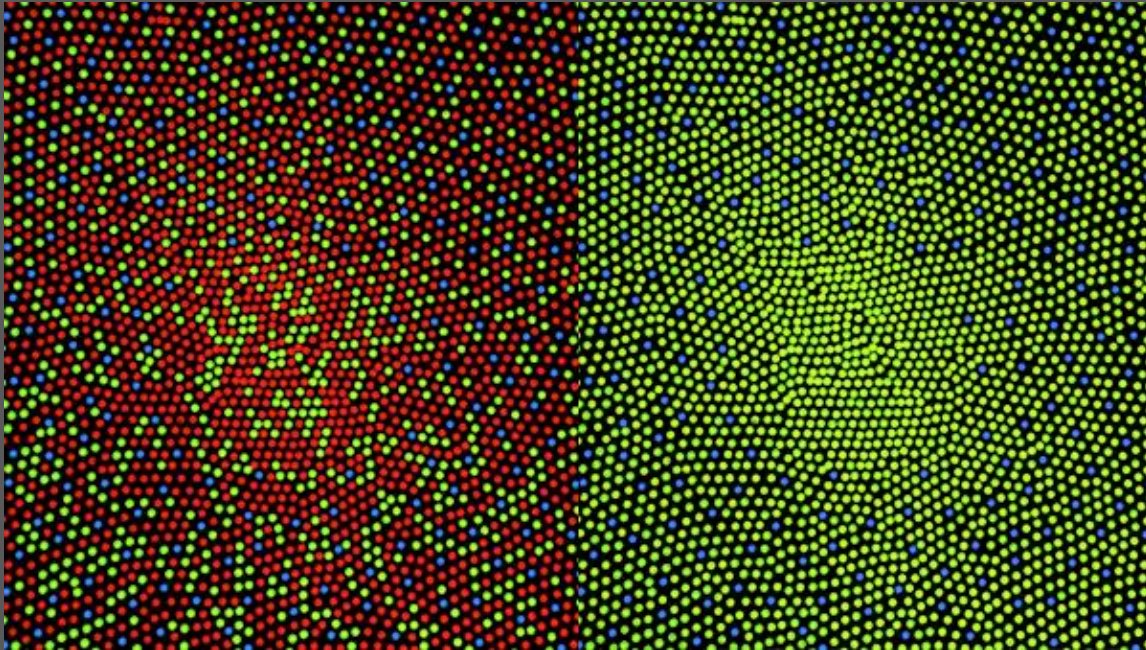


Standard Observer (1931)

Chromaticity diagram (1931)



Colour vision variance



NORMAL VISION



DEUTERANOMALY



PROTANOPIA

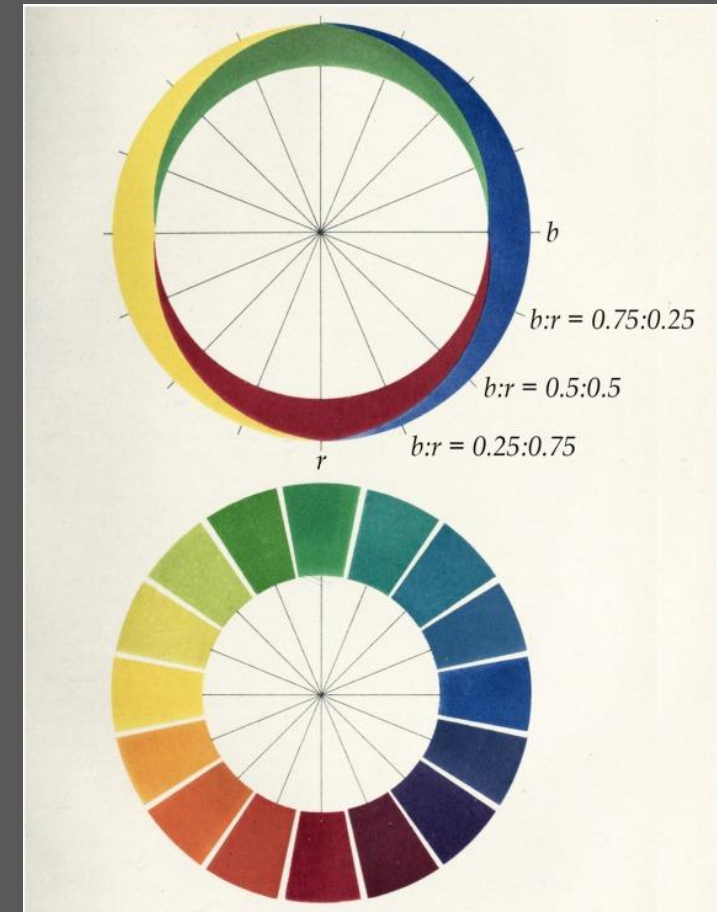
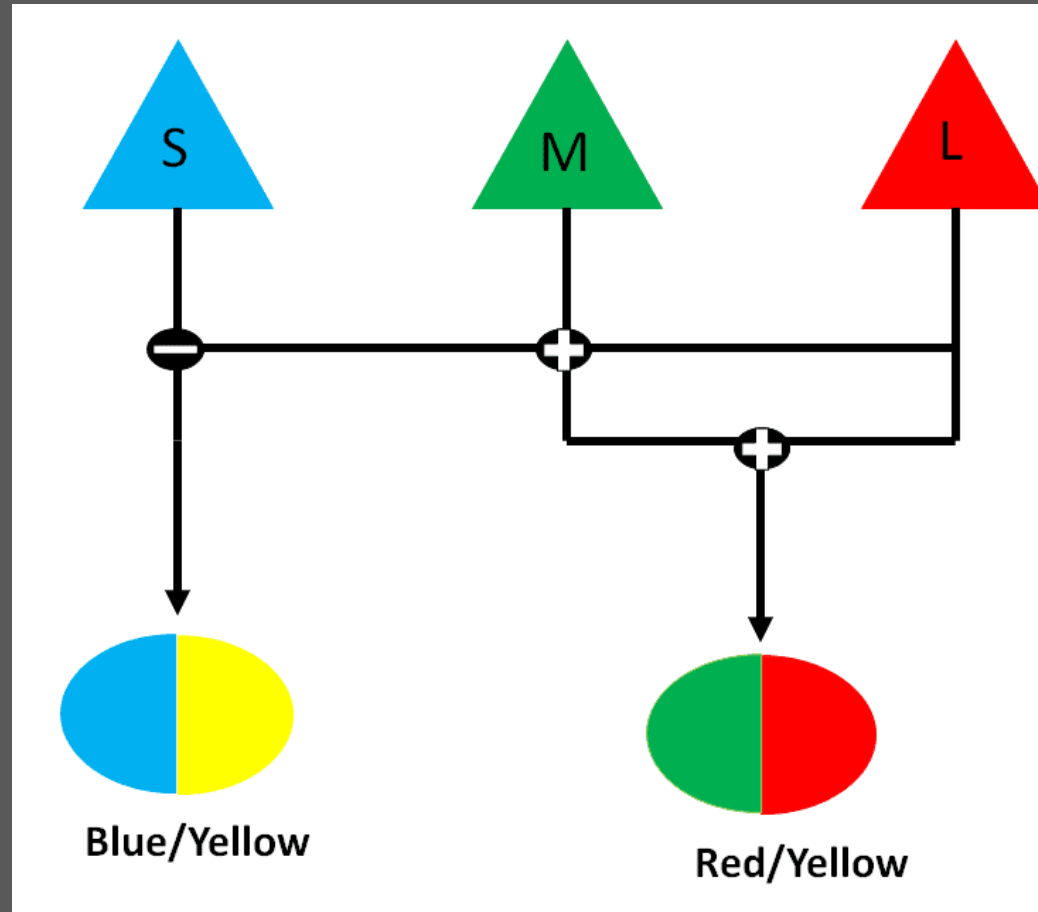


TRITANOPIA



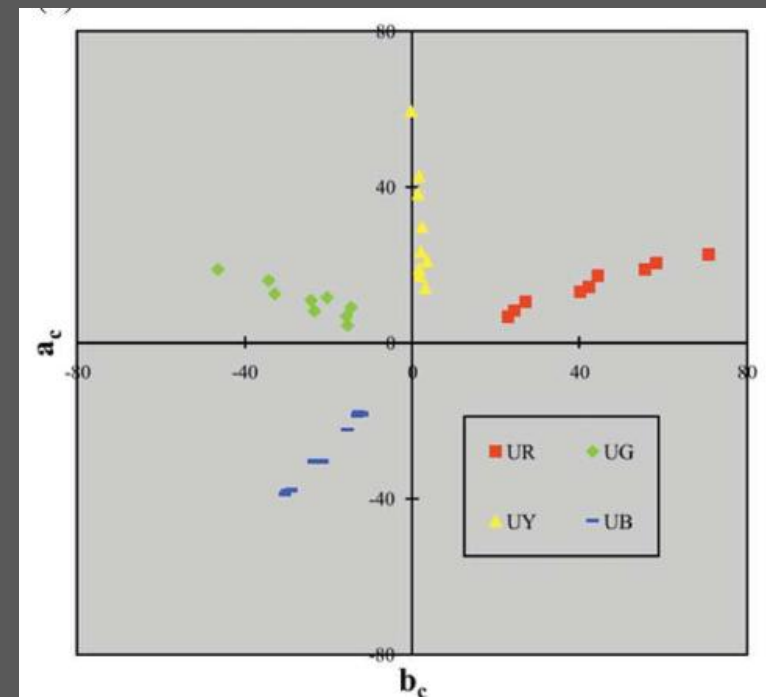
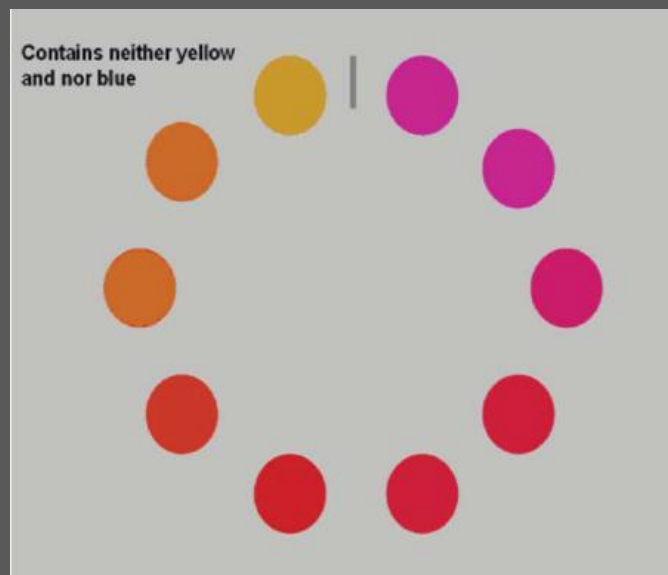
<https://www.color-blindness.com/coblis-color-blindness-simulator/>

Primaries and opponent colour theory



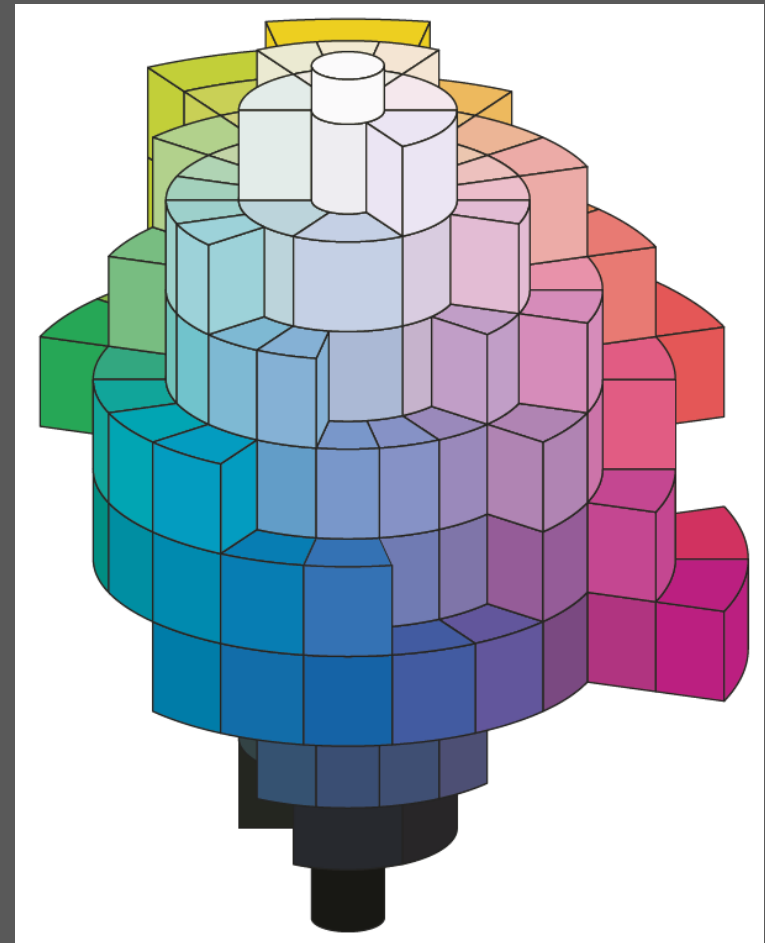
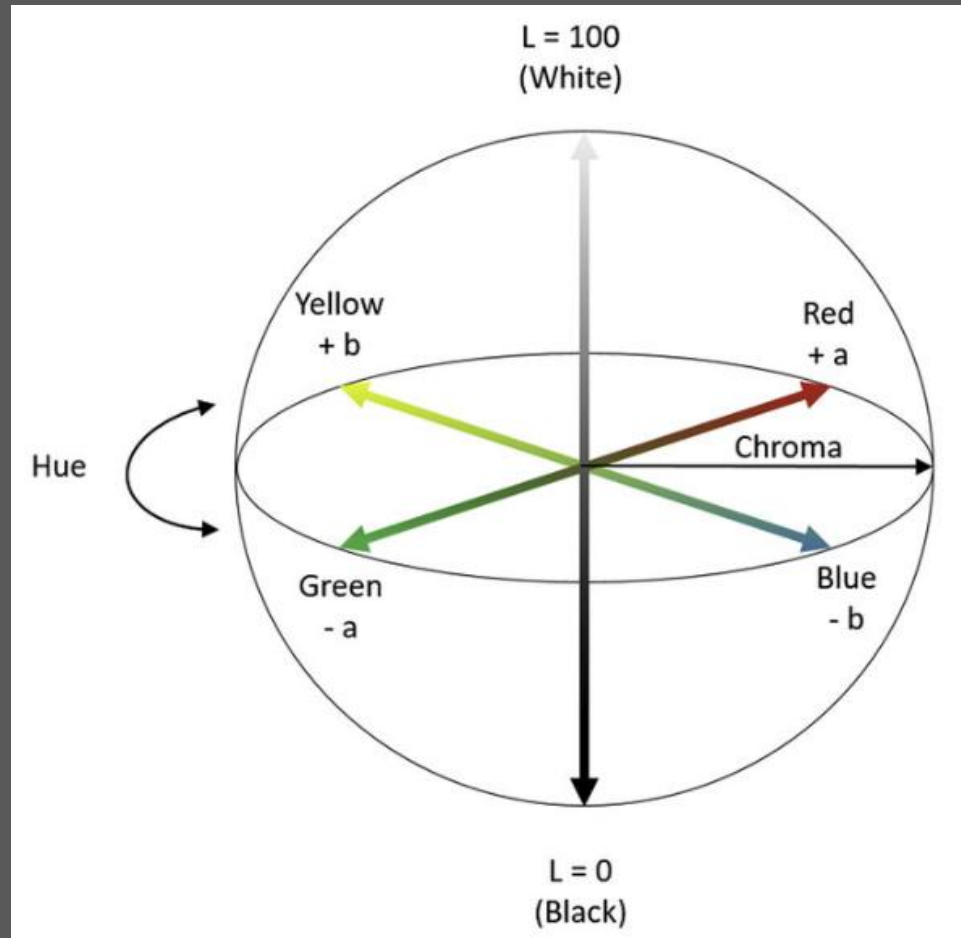
Hering (1878/1964)

Unique hues and colour appearance



Xiao, Fu, Mylonas, Karatzas & Wuerger (2011)

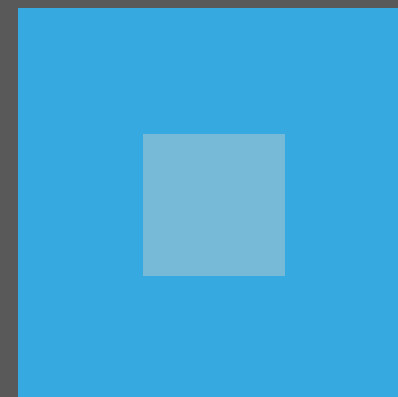
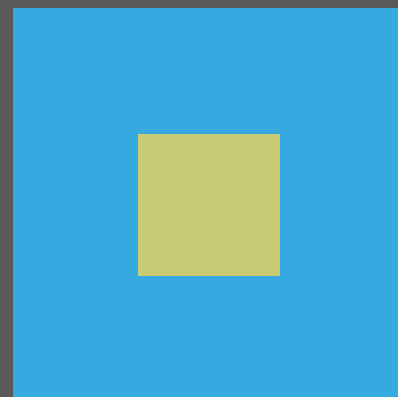
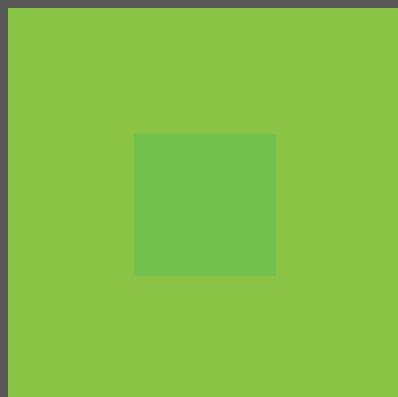
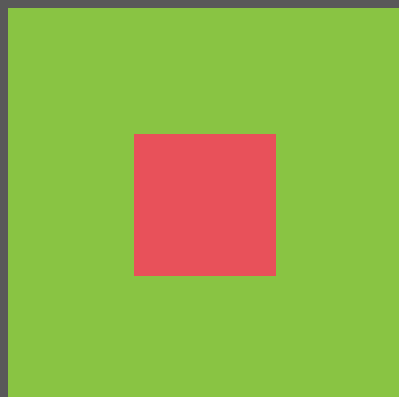
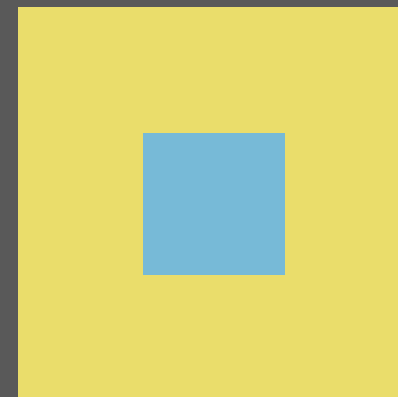
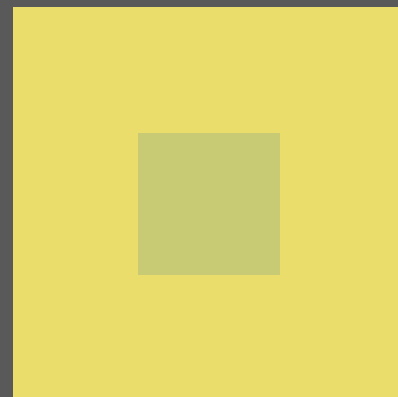
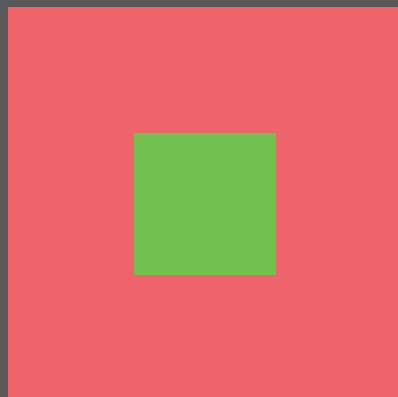
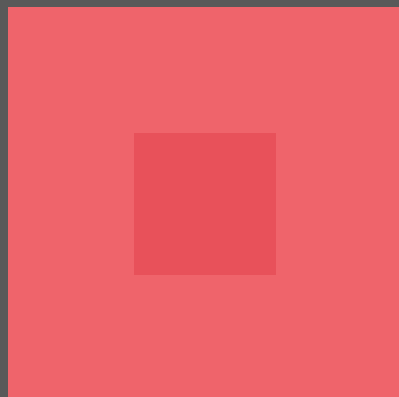
CIE LAB and CIE LCh colour spaces



“In order to use colour effectively it is necessary to recognize that it deceives continually”

Josef Albers, Interaction of Color

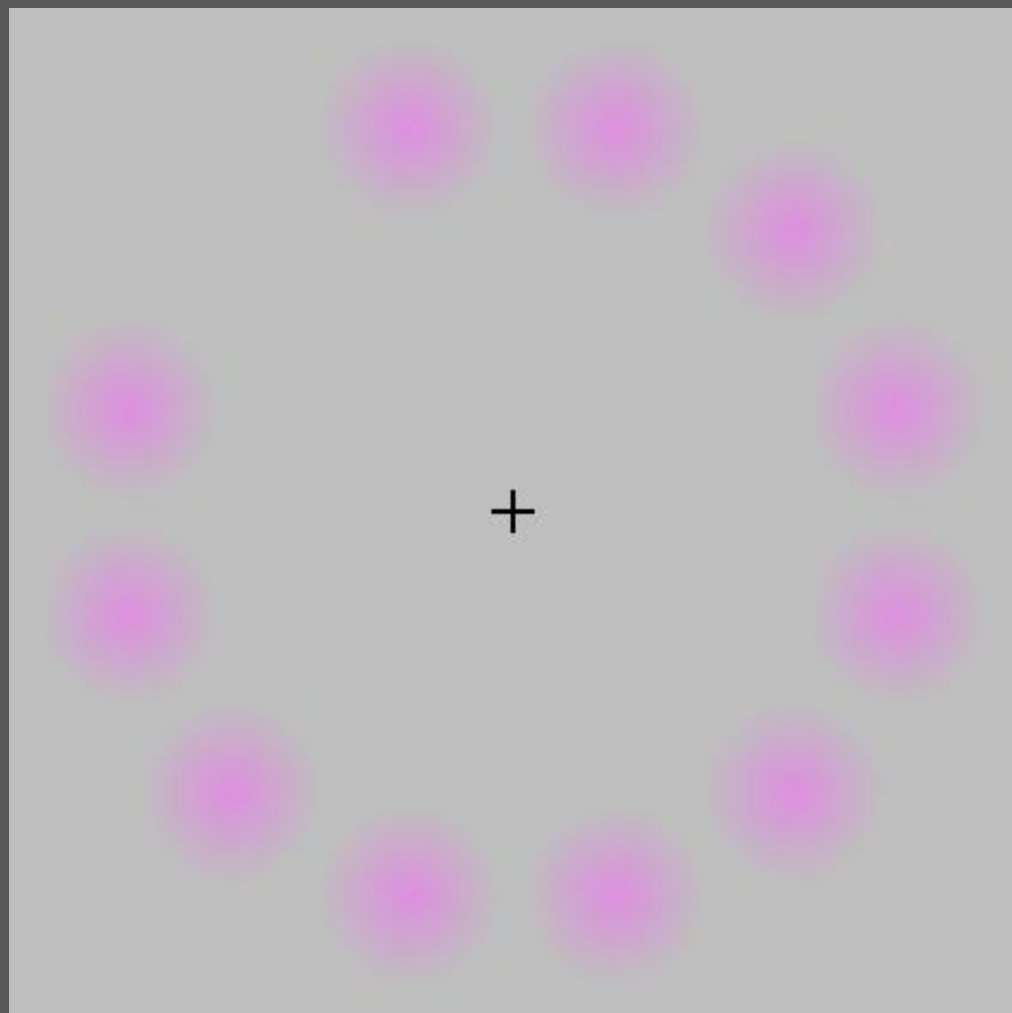
Appearance: colour contrast



Appearance: colour contrast

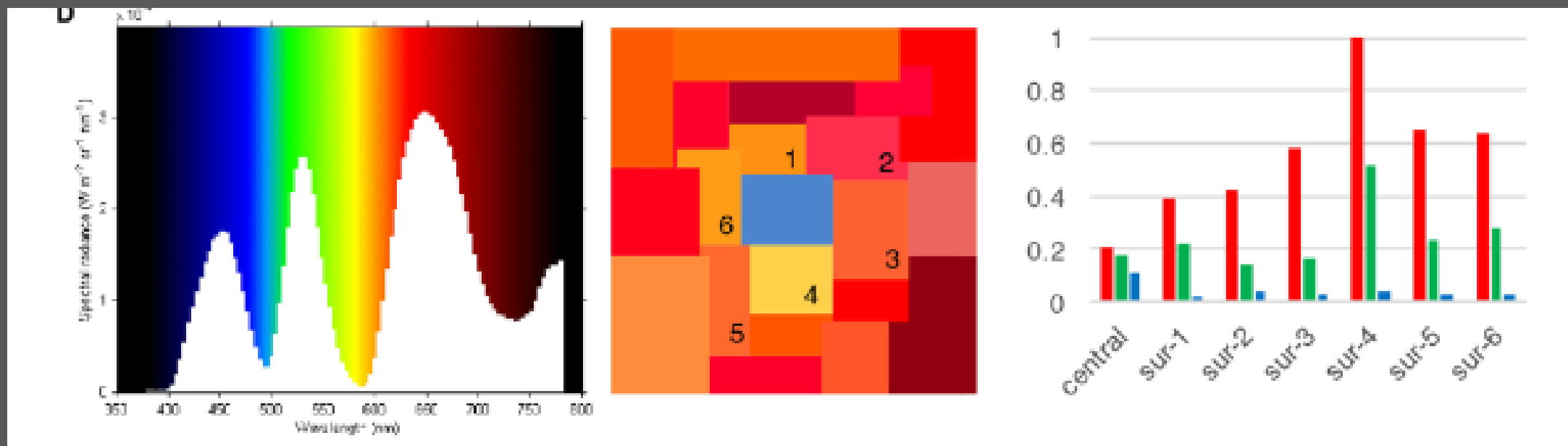


Appearance: colour after images



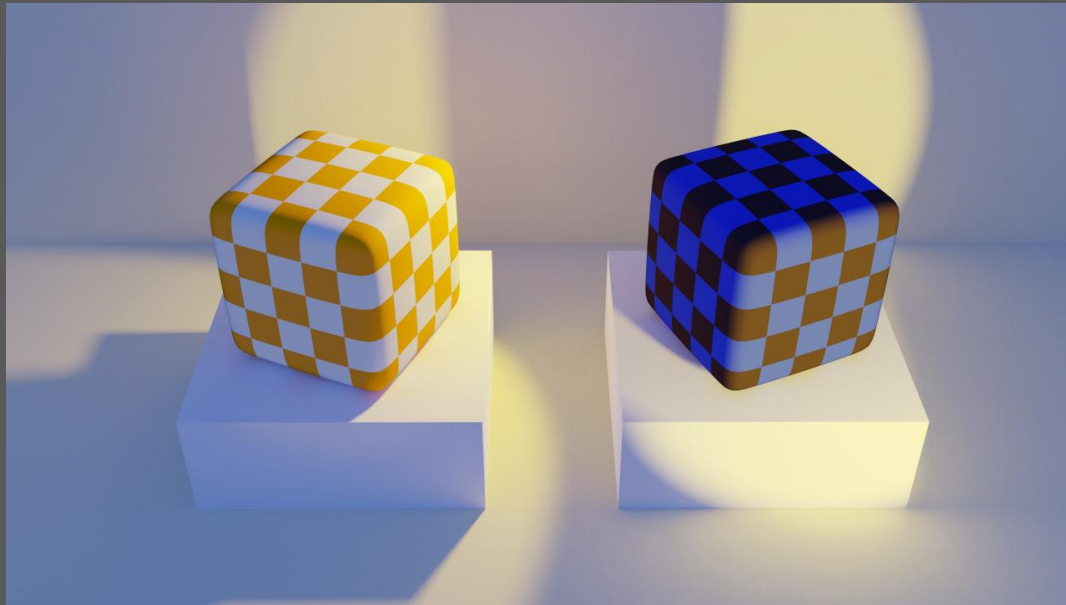
Lilac Chaser: Hinton (2005)

Appearance: Mondrian colour after images

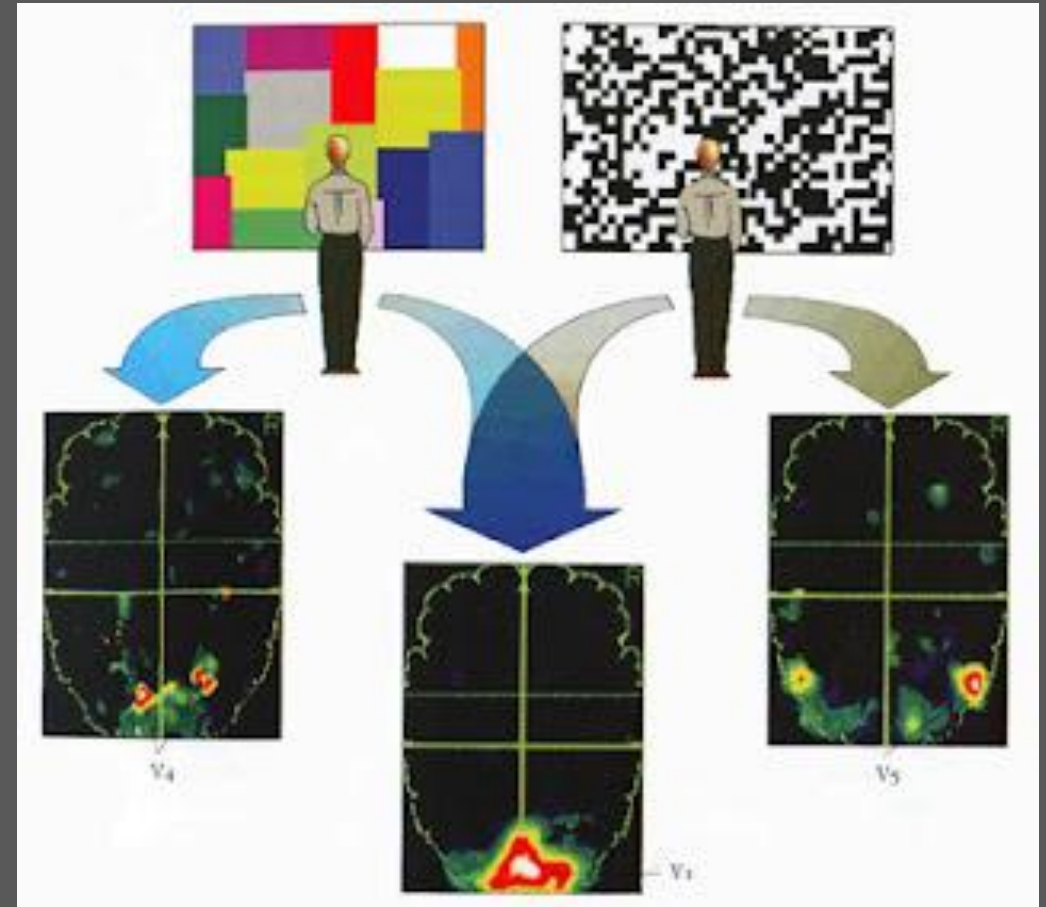


Zeki, Cheadle, Pepper & Mylonas (2017)

Colour constancy

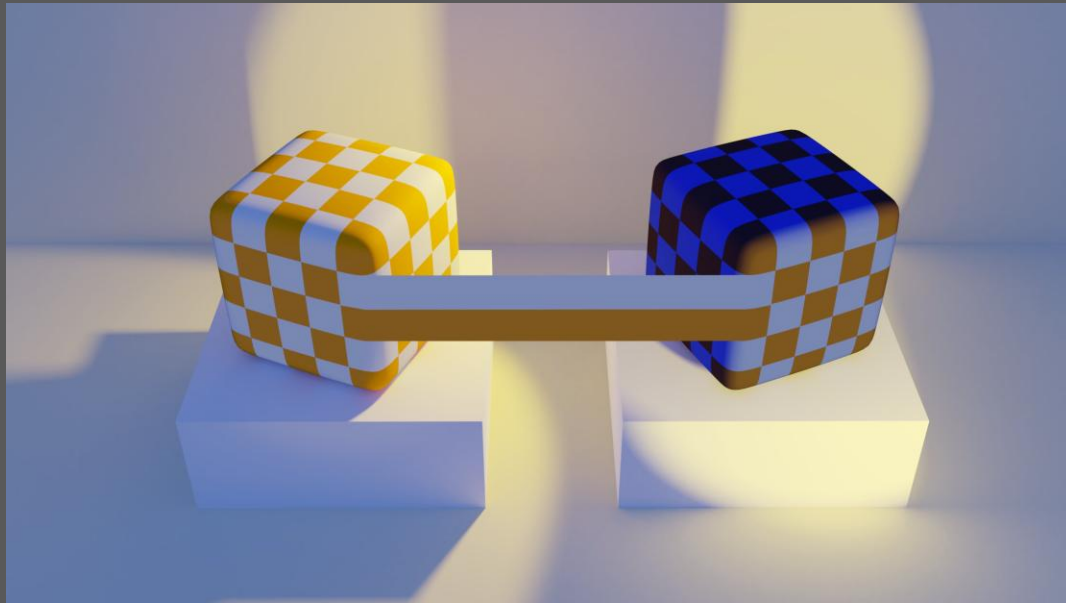


Cubes illusion, Vladusich (2021)

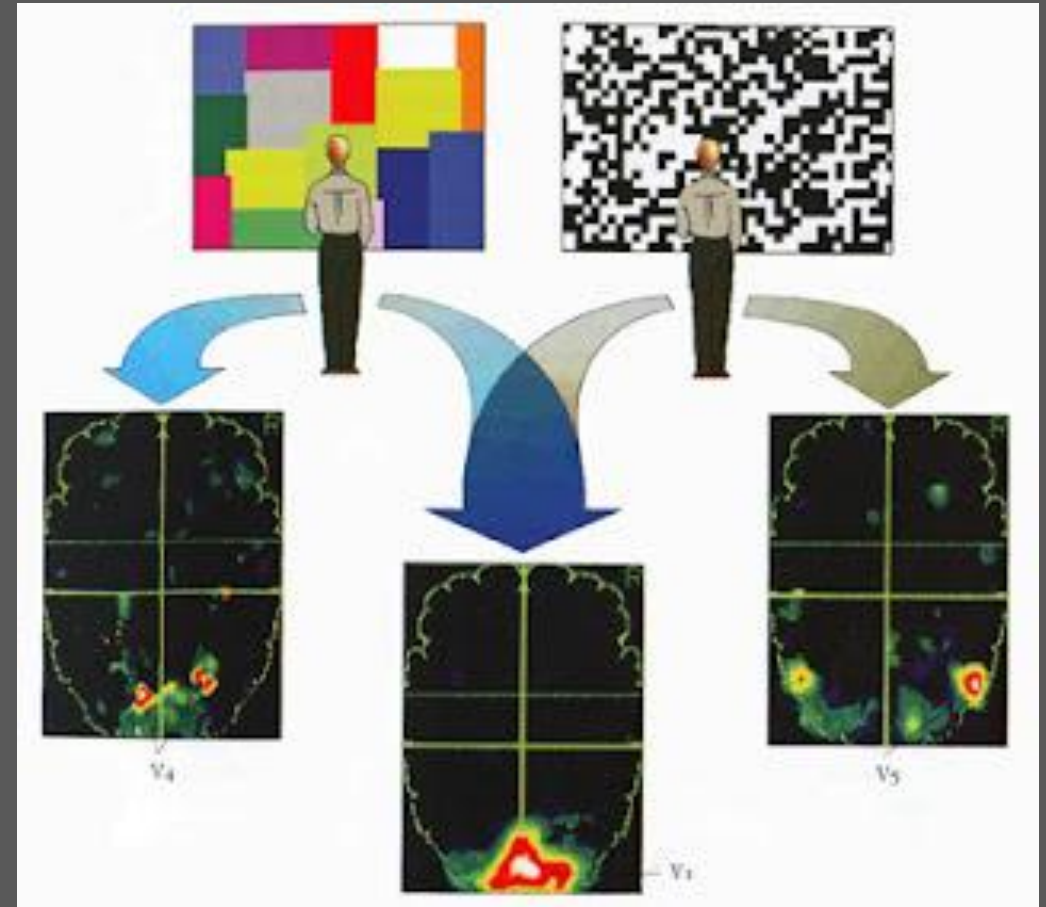


Brain activations for visual stimuli, Zeki (1993)

Colour constancy

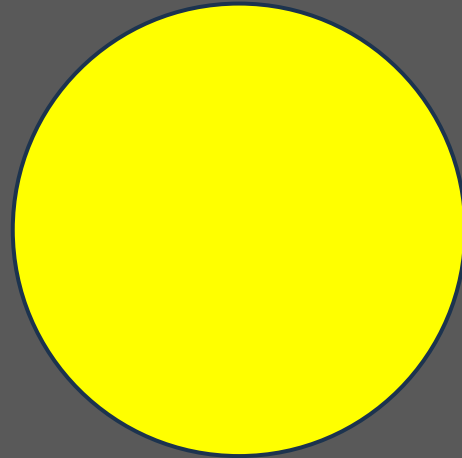


Cubes illusion, Vladusich (2021)

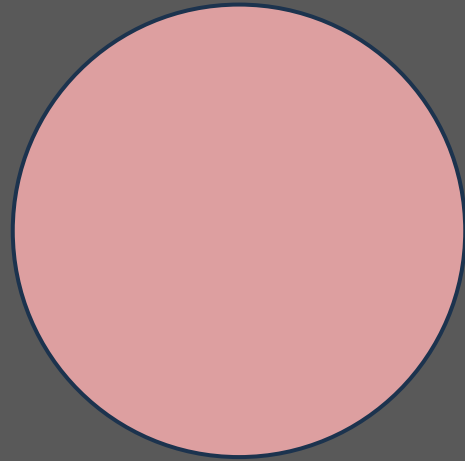


Brain activations for visual stimuli, Zeki (1993)

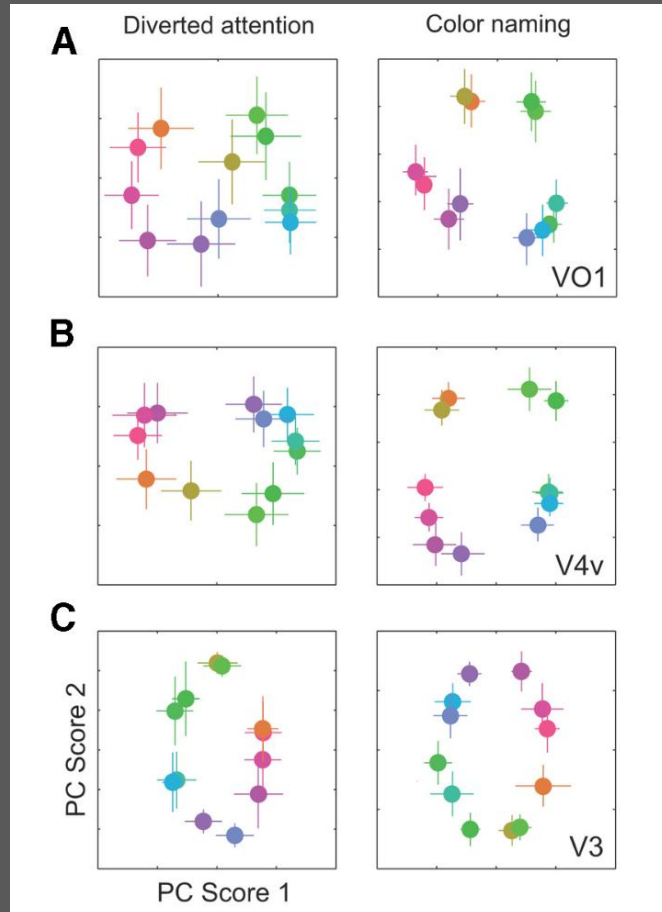
Colour cognition: name the colour?



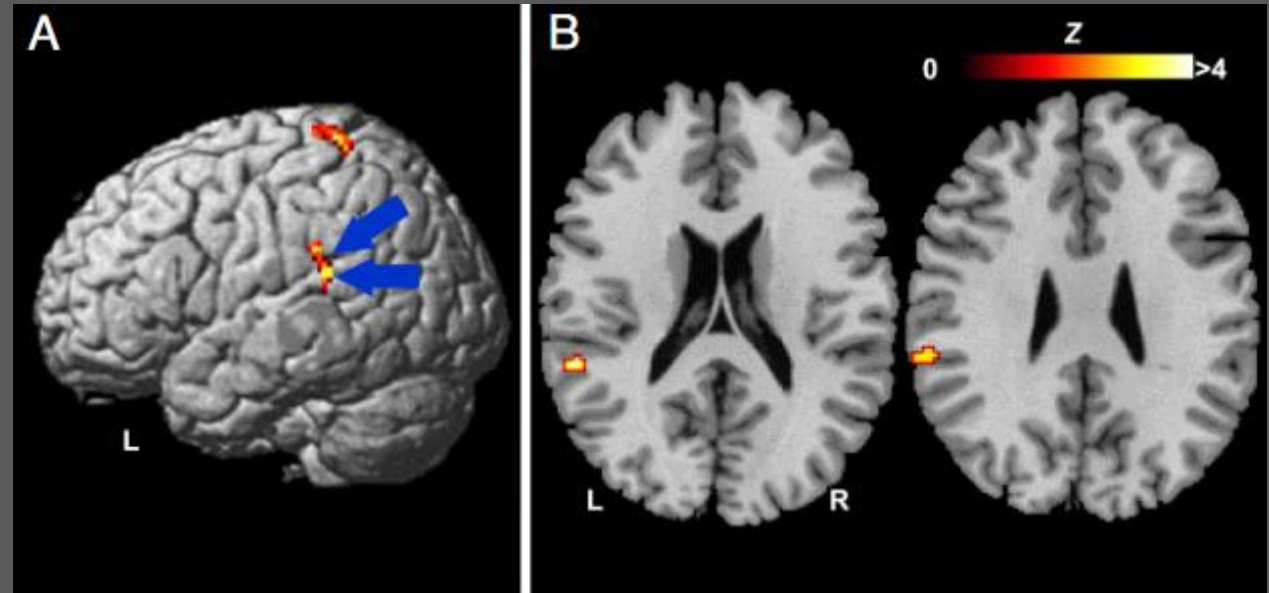
Colour cognition: name the colour?



Colour naming in the brain

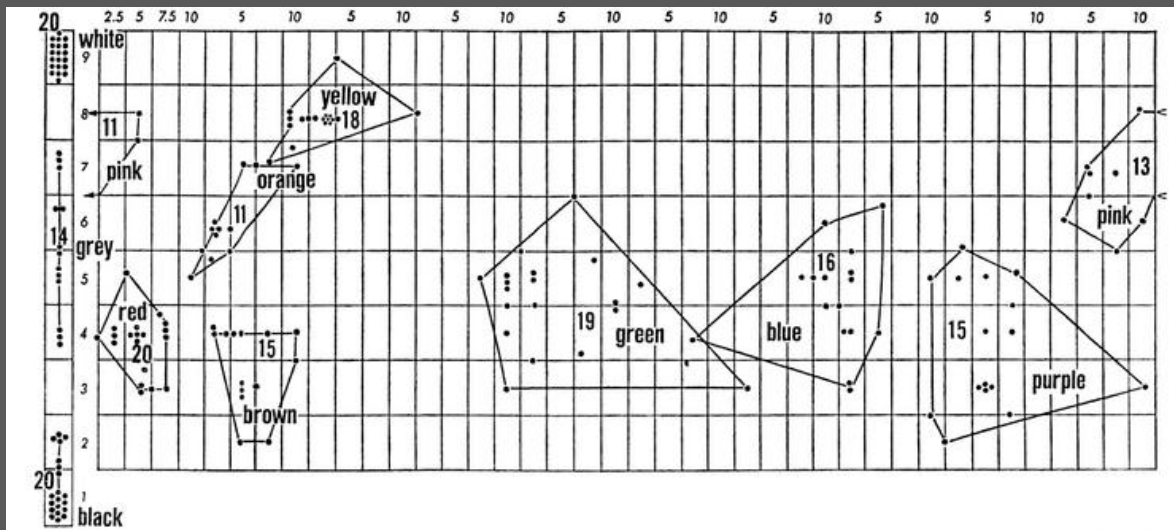
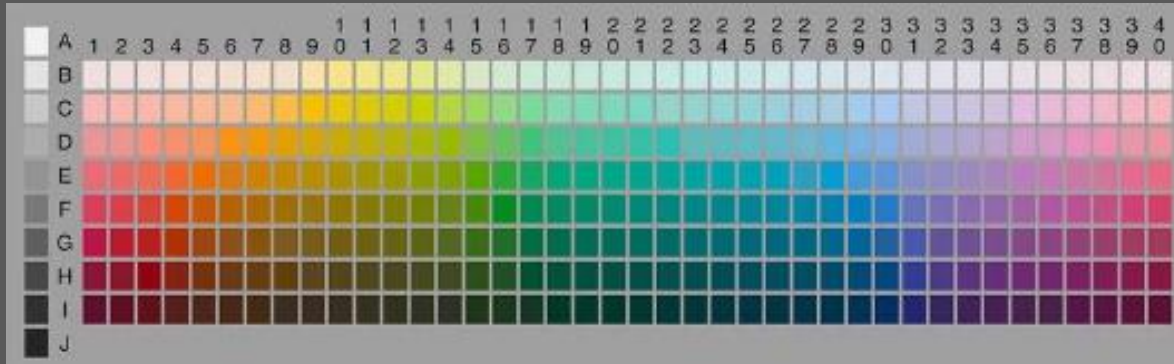


Brouwer & Heeger (2013)



Tan et al. (2008)

Basic colour terms



[white
black] < [red] < [green
yellow] < [blue] < [brown] < [purple
pink
orange
grey]

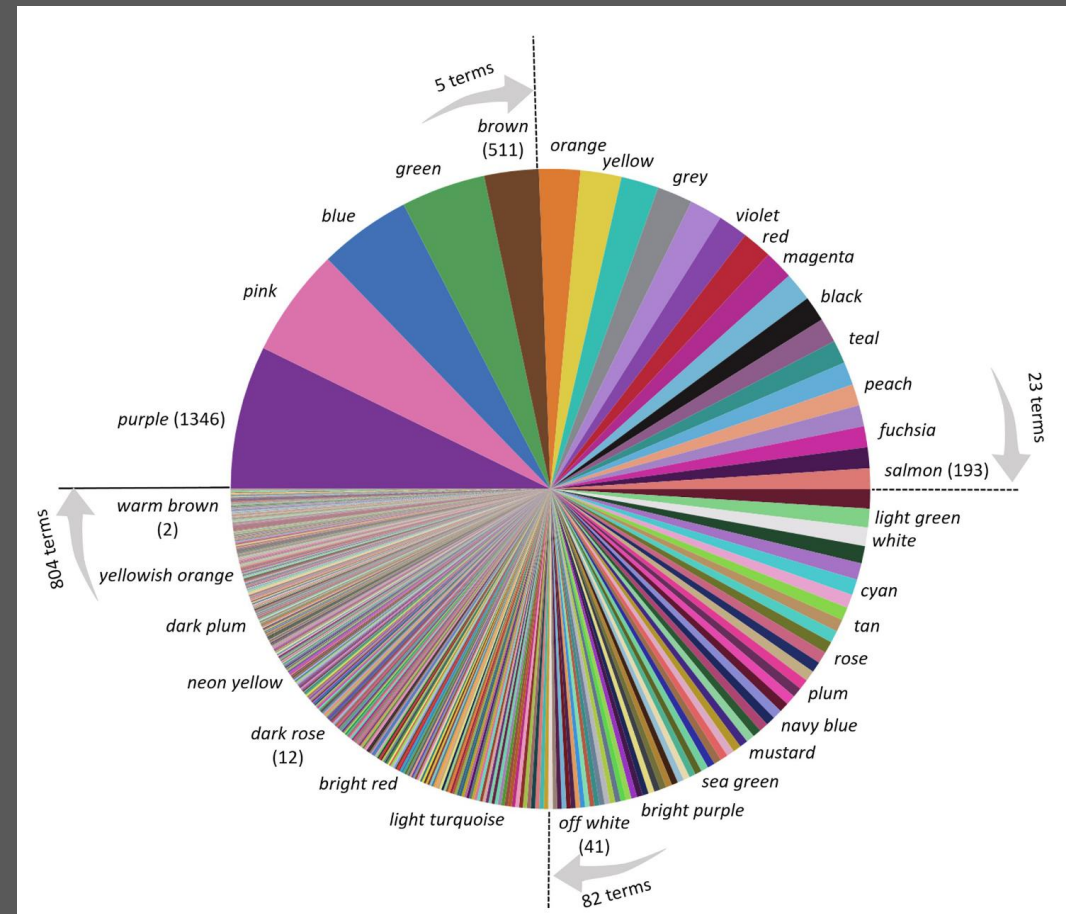
Berlin & Kay (1969/1991)

An online colour naming experiment



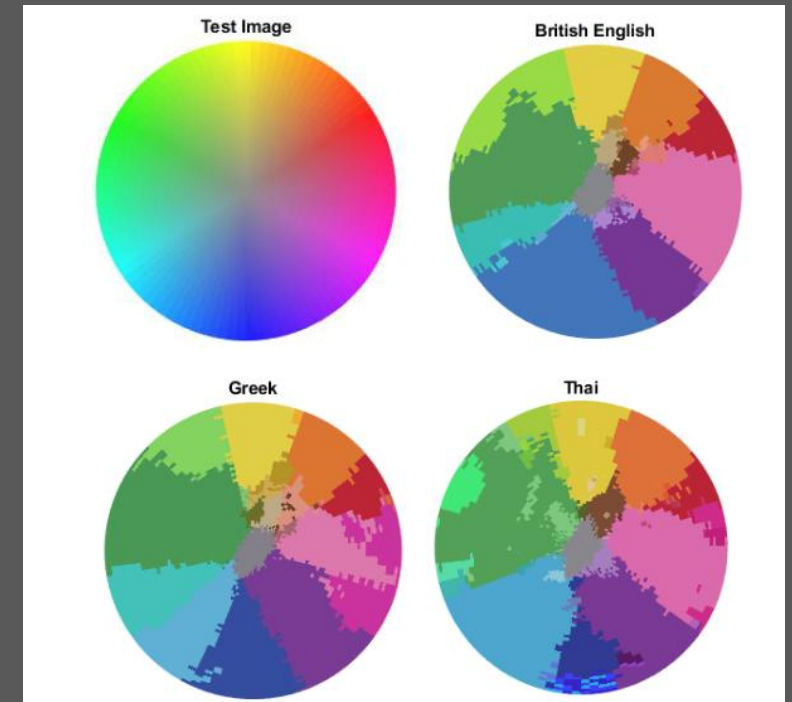
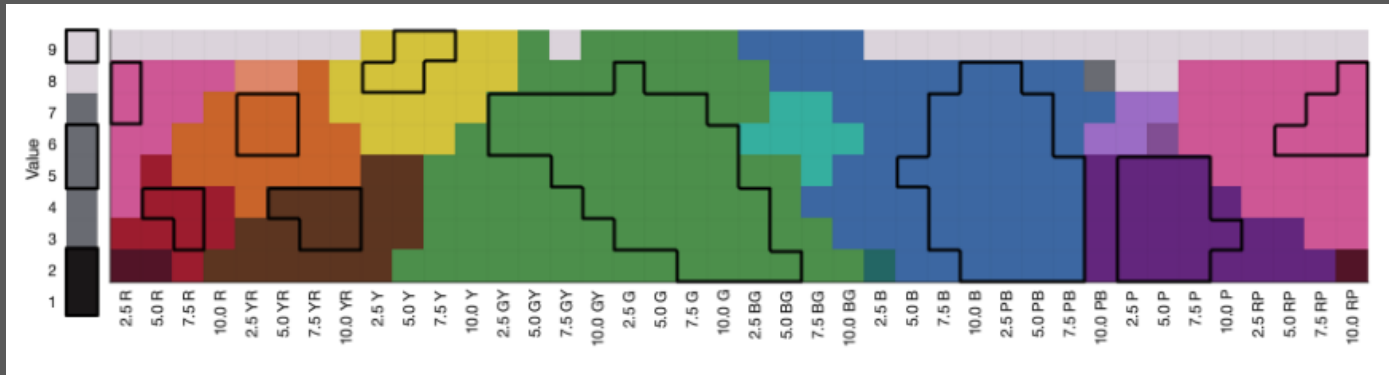
Mylonas & MacDonald (2010)

Griffin and Mylonas (2019)

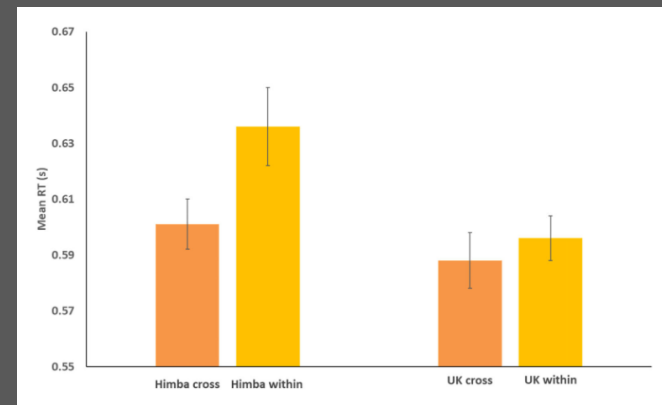
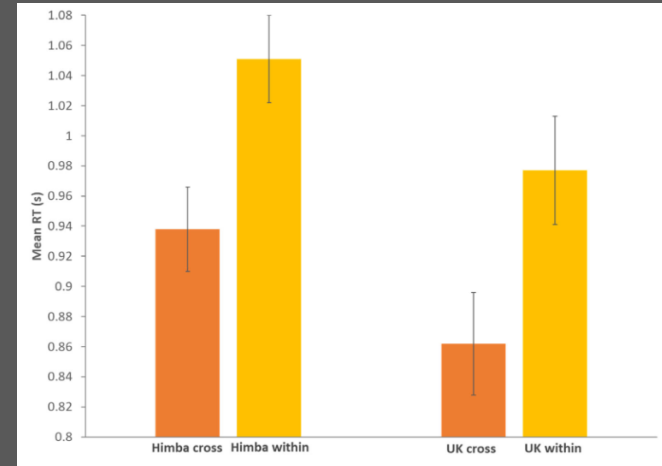
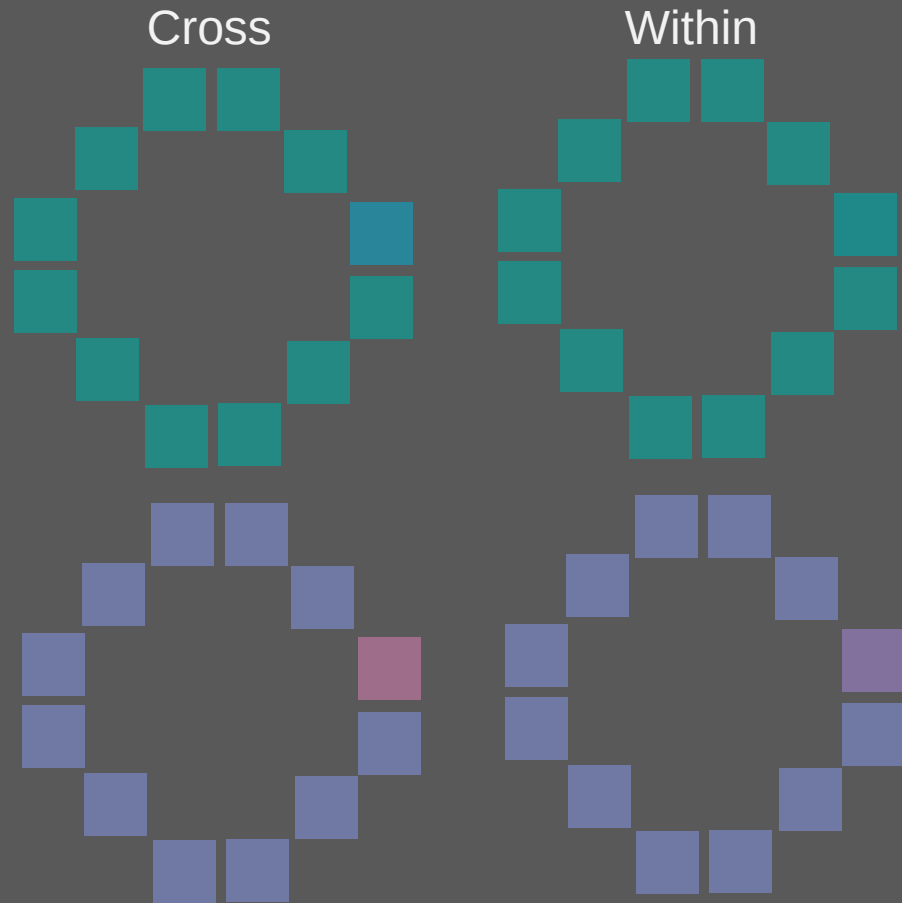


Multilingual color naming model

Surface colours classified in English



Naming influences perception



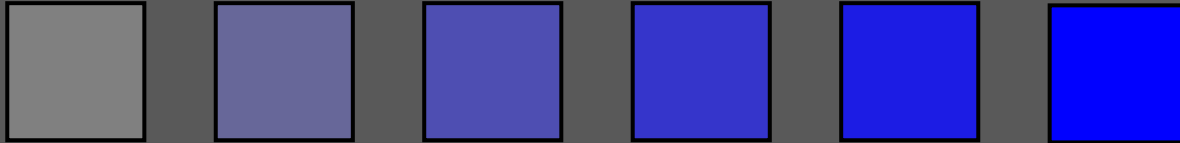
Wakui, Mylonas, Caparos & Davidoff (2025)

Colour dimension orders

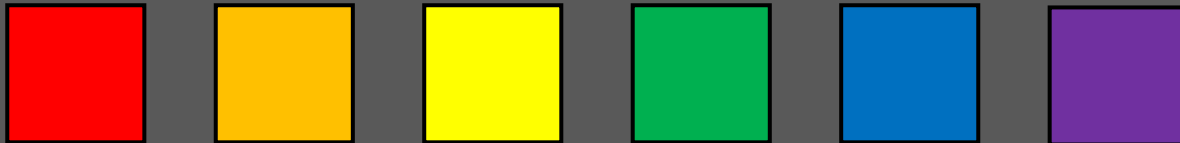
- Lightness



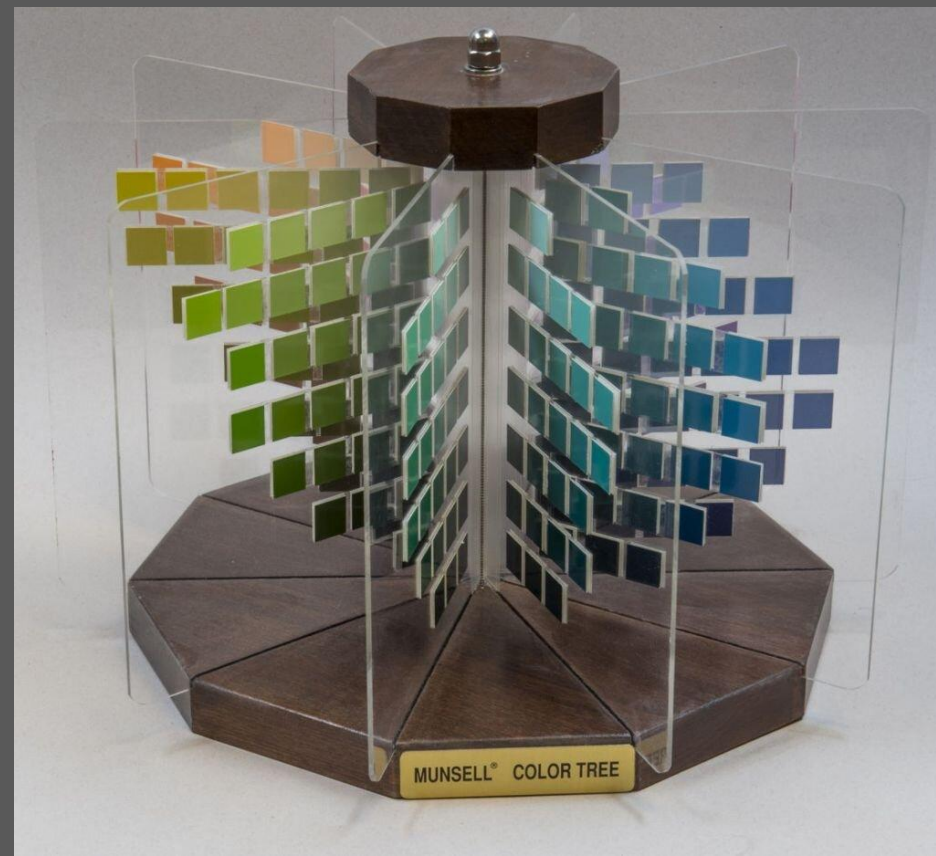
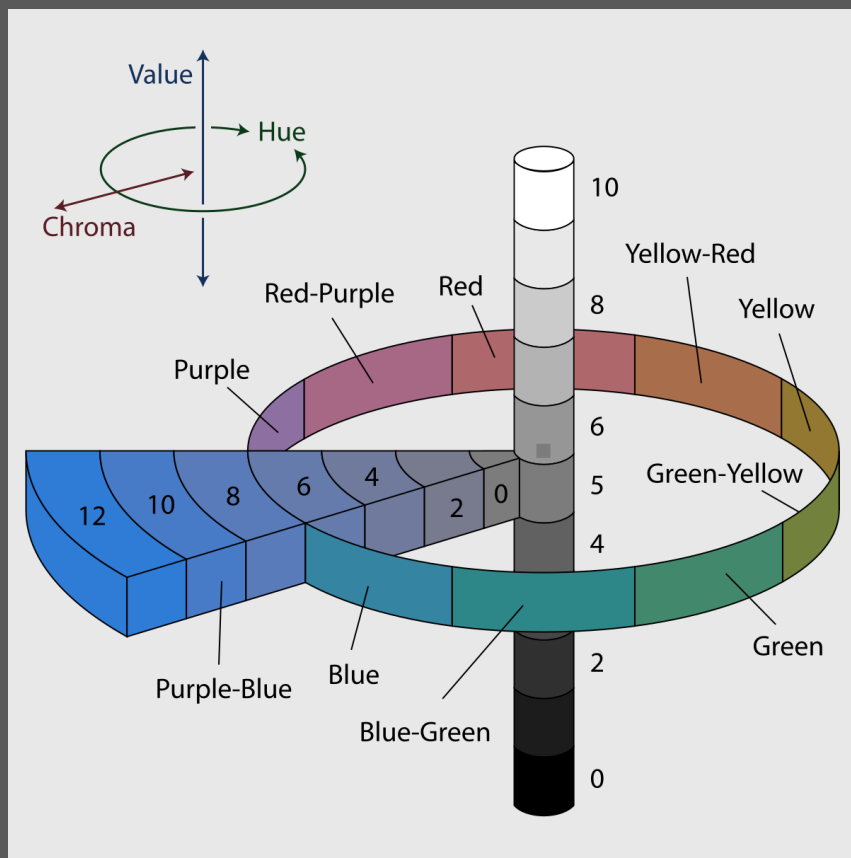
- Saturation



- Hue



The Munsell colour system



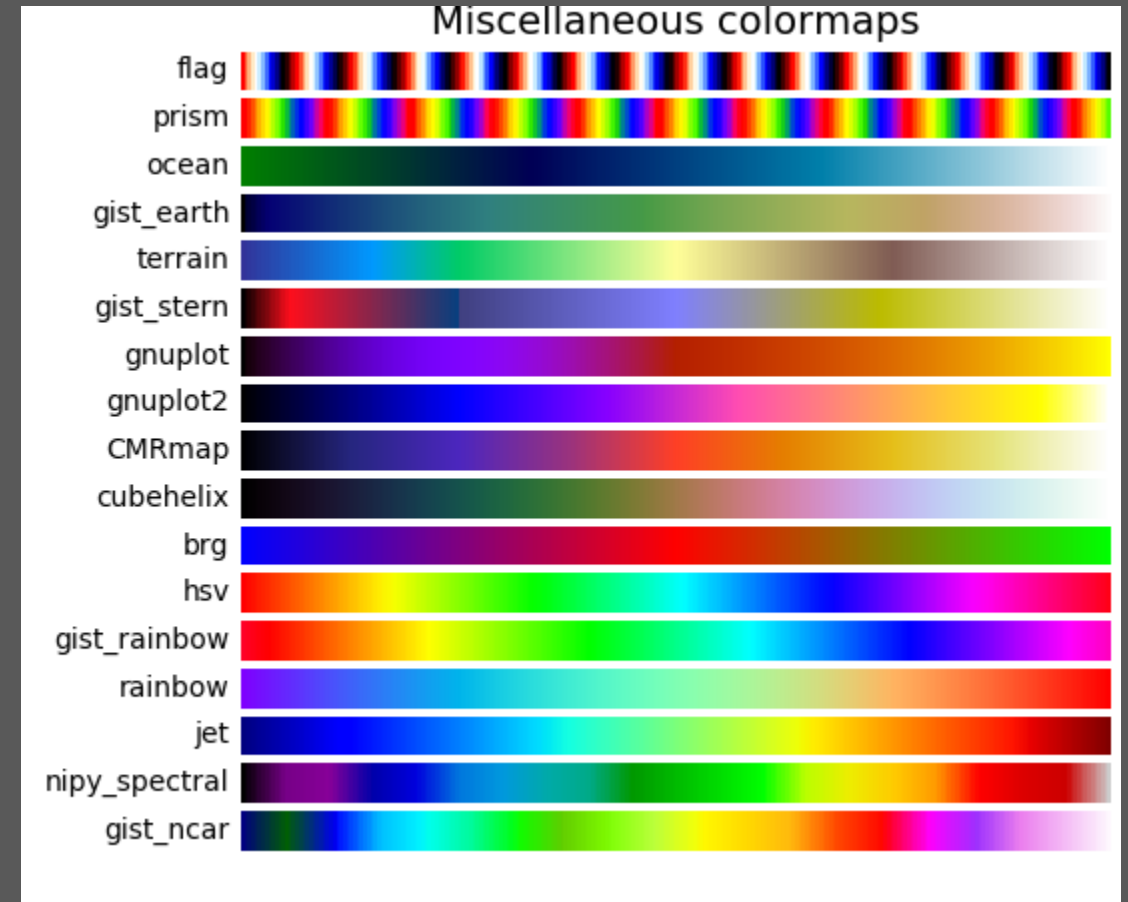
Munsell (1905)

Colour encoding

Maps, palettes, rules of thumb

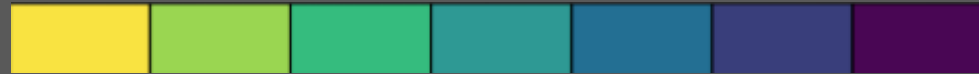
Colour maps

- Enhances Data Interpretation
- Improves Accessibility
- Supports Effective Storytelling
- Maintains Consistency

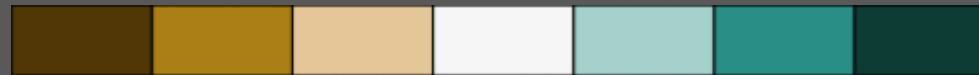


Main types of colour maps

- **Sequential:** Uses a gradual shift in lightness or saturation or hue to represent ordered data (e.g., temperature maps)



- **Diverging:** Features two contrasting colours meeting at a neutral midpoint, ideal for highlighting deviations (e.g., profit/loss).



- **Categorical:** Assigns distinct colours to separate groups, best for classifying discrete categories (e.g., political maps).



Rainbow colour maps?

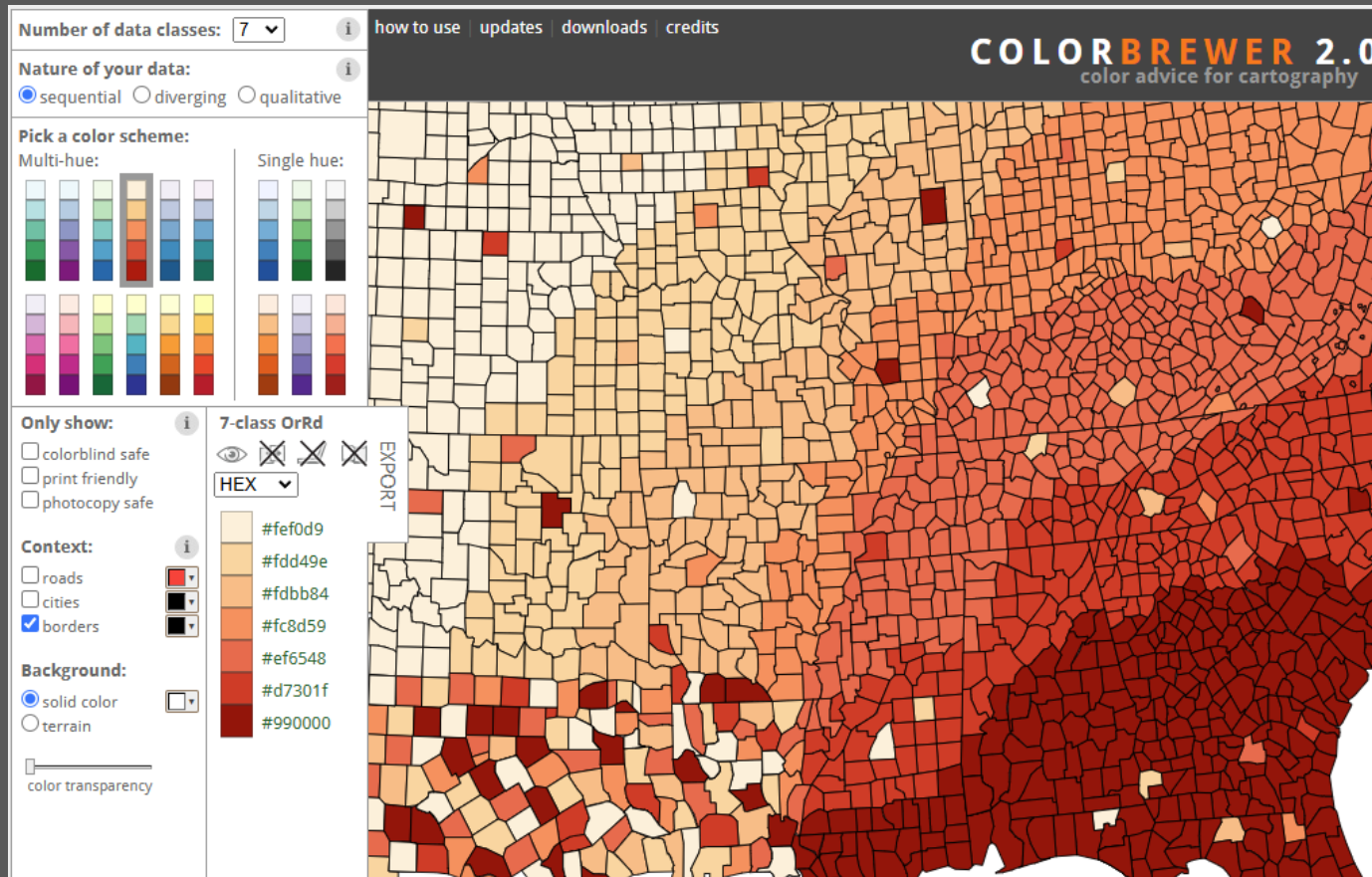


<https://hellerweather.com/data-visualization-and-the-overused-rainbow-color-table/>

Designing a colour map

- Use perceptually distinct colours.
- Maintain alignment between colours and concepts.
- Ensure aesthetic appeal and clarity.
- Consider colour vision deficiencies.
- Retain visibility when converted to black & white.
- Keep it simple by avoiding overload!

Colour tools: colorbrewer



<https://colorbrewer2.org/>

Colour tools: colorgorical

Colorgorical Source

Generate



Number of colors

5

Score importance

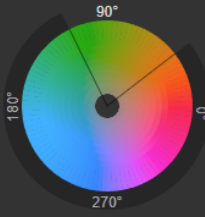
Perceptual Distance

Name Difference

Pair Preference

Name Uniqueness

Select hue filters



Drag wheel, or add angle:

Results: Color space Hex RGB Lab LCH Array format " ' No quote Charts     Clear all

["Lab(70,-45,30)", "Lab(30,-5,0)", "Lab(75,30,-20)", "Lab(35,-40,35)"]

Lab(70,-45,30) + start

Lab(30,-5,0) + start

Lab(75,30,-20) + start

Lab(35,-40,35) + start

["Lab(85,-15,-15)", "Lab(40,-20,-15)", "Lab(85,-45,0)", "Lab(30,10,-40)"]

Lab(85,-15,-15) + start

Lab(40,-20,-15) + start

Lab(85,-45,0) + start

Lab(30,10,-40) + start

Instructions

To generate a palette with n colors, just enter the number of colors you want and click *Generate*. Bigger palettes will take longer than smaller palettes to make. Results will automatically appear when ready.

About

Colorgorical was built by Connor Gramazio with advisement from David Laidlaw and Karen Schloss.

Documentation

<http://vrl.cs.brown.edu/color>

Colour tools: colour palette analyser

Color Name Distance										Saliency	Name
0.00	1.00	1.00	0.89	0.08	1.00	0.19	1.00	1.00	0.88	.44	blue 61.5%
1.00	0.00	0.99	1.00	1.00	0.81	1.00	0.78	1.00	0.99	.21	red 21.1%
1.00	0.99	0.00	1.00	0.98	0.99	1.00	1.00	0.10	1.00	.39	green 42.8%
0.89	1.00	1.00	0.00	0.92	1.00	0.80	0.84	1.00	0.31	.42	purple 57.8%
0.08	1.00	0.98	0.92	0.00	1.00	0.21	1.00	0.97	0.88	.24	blue 40.4%
1.00	0.81	0.99	1.00	1.00	0.00	1.00	0.92	1.00	1.00	.28	orange 36.3%
0.19	1.00	1.00	0.80	0.21	1.00	0.00	0.94	0.97	0.58	.16	blue 25.6%
1.00	0.78	1.00	0.84	1.00	0.92	0.94	0.00	0.99	0.76	.10	pink 21.8%
1.00	1.00	0.10	1.00	0.97	1.00	0.97	0.99	0.00	0.96	.21	green 30.8%
0.88	0.99	1.00	0.31	0.88	1.00	0.58	0.76	0.96	0.00	.25	purple 22.7%
Excel-10										Average	0.86 .27

Color Name Distance										Saliency	Name
0.00	1.00	1.00	1.00	0.96	1.00	1.00	0.99	1.00	0.19	.47	blue 65.3%
1.00	0.00	1.00	0.98	1.00	1.00	1.00	1.00	0.97	1.00	.87	orange 92.2%
1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.70	0.99	.70	green 81.3%
1.00	0.98	1.00	0.00	1.00	0.96	0.99	1.00	1.00	1.00	.64	red 79.3%
0.96	1.00	1.00	1.00	0.00	0.95	0.83	0.98	1.00	0.97	.43	purple 52.5%
1.00	1.00	1.00	0.96	0.95	0.00	0.99	0.96	0.96	1.00	.47	brown 60.5%
1.00	1.00	1.00	0.99	0.83	0.99	0.00	1.00	1.00	1.00	.47	pink 60.3%
0.99	1.00	1.00	1.00	0.98	0.96	1.00	0.00	1.00	0.99	.74	grey 83.7%
1.00	0.97	0.70	1.00	1.00	0.96	1.00	1.00	0.00	1.00	.11	yellow 20.1%
0.19	1.00	0.99	1.00	0.97	1.00	1.00	0.99	1.00	0.00	.25	blue 27.2%
Tableau-10										Average	0.96 .52

<http://vis.stanford.edu/color-names>

Colour design guidelines

- Ensure Clarity & Differentiation
- Consider Audience & Accessibility
- Use Research-Based Data visualisation Tools
- Enhance Communication & Memorability
- Balance Aesthetics & Functionality

Questions?